

What Drives First-Year MLB Arbitration Salaries for Starting Pitchers?

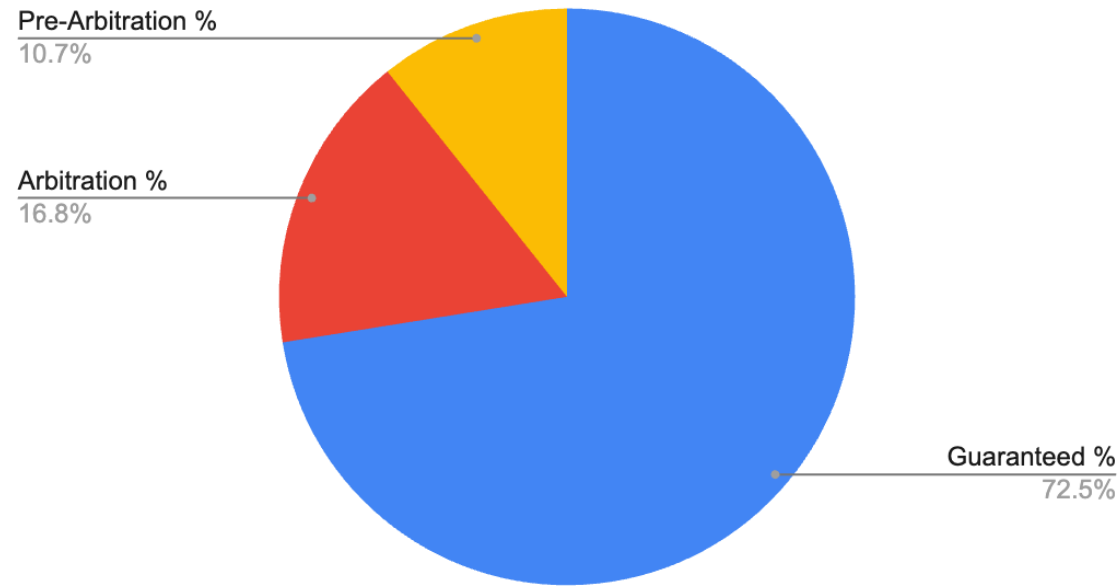
A Fangraphs + Spotrac + Statcast Econometric Analysis (2016-2026)

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Why This Matters?

- MLB arbitration affects millions in payroll decisions. Teams and players argue using performance metrics, but which metrics influence first-year salaries?

Average MLB Payroll Composition (2026)



Data Obtained via Fangraphs Payroll Breakdown, Author Calculations

Data

Component	Source
Arbitration Salaries	Spotrac
Pitching Performance	Statcast
Sample	Arbitration 1 + Super Two SP
Years	2016-2026
Observations	124

Source: Spotrac, Fangraphs, Statcast, Author calculations

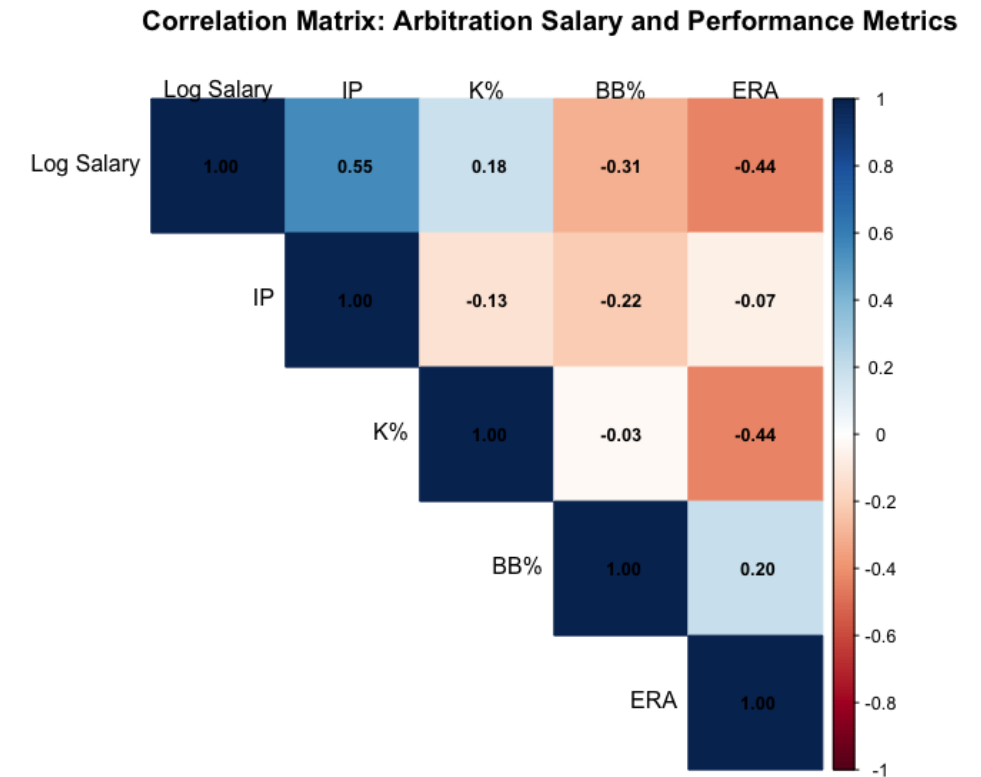
Descriptive Statistics

Variable	Mean	SD	Min	Max
Salary	\$2.89M	\$1.09M	\$0.88M	\$7.25M
IP	132	43	51	232
K%	22.8	4.81	12.6	35.6
BB%	7.6	1.8	3	13.5
ERA	3.96	0.97	1.78	6.73

- Salary distributions are heavily right-skewed, motivating a logarithmic specification

Correlation Heatmap

- Innings Pitched and Log Salary exhibit the strongest positive relationship



Model Specification

$$\log(\text{Salary}_i) = \beta_0 + \beta_1(IP / 10)_i + \beta_2 K_i + \beta_3 BB_i + \beta_4 ERA_i + \delta_t + u_i$$

- Salary modeled in logs
- Year fixed effects control for differences in salary environments across arbitration classes (2016-2026)
- Prior-year performance predicts arbitration outcomes

δ_t = Year fixed effects

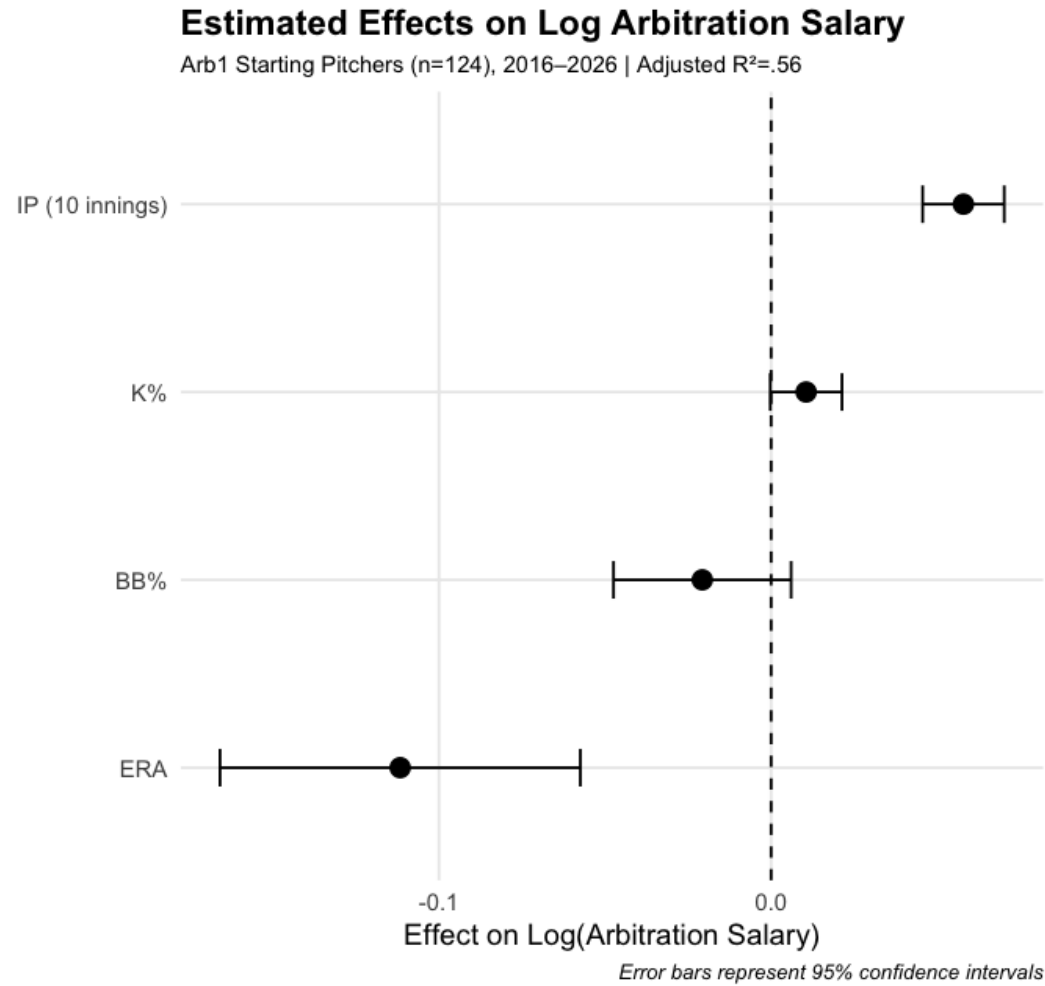
u_i = error term

Results

Adjusted $R^2 = .56$
N = 124

Variable	Effect	p-value
IP (10 Innings)	+5.8%	< .001
K%	+1.05%	.055
BB%	-2.1%	.127
ERA	-11.2%	< .001

Coefficient Plot



Source: Spotrac, Fangraphs, Statcast, Author calculations

Baseball Interpretation

- Key Findings

Estimated average effects holding performance and season constant

- Workload dominates
 - 10 additional innings pitched → ~ 5.8% higher salary
- ERA still matters
 - One-run ERA increase → ~ 11% lower salary
- Strikeouts help
 - Moderate evidence
- Walks appear less influential
- Arbitration may reward traditional workload and run-prevention metrics more heavily than advanced indicators

Model Successes and Failures

Largest Underpredictions

Pitcher	Season	Predicted	Actual	Difference
Shane Bieber	2022	\$2.65M	\$6.0M	+\$3.35M
Corbin Burnes	2022	\$4.74M	\$6.5M	+\$1.76M
Hunter Brown	2023	\$4.35M	\$5.71M	+\$1.36M

- Interpretation
 - The largest underpredictions occur among elite pitchers, suggesting arbitration awards may incorporate factors not fully captured by traditional performance metrics

Model Successes and Failures

Largest Overpredictions

Pitcher	Season	Predicted	Actual	Difference
Michael King	2023	\$2.09M	\$1.3M	-\$788K
Cal Quantrill	2022	\$3.26M	\$2.51	-\$752K

- Interpretation
 - Performance metrics alone may not capture role or context

Limitations

- Possible omitted variables
 - WAR
 - Awards
 - Injuries
 - Reputation
 - Role
 - Comparable-player effects
 - Advanced Statcast metrics excluded from baseline specification
- Violations of:
 $E(u | X) = 0 \rightarrow$ may create omitted variable bias

Conclusion

- Workload and run prevention significantly influence first-year arbitration salaries
- Arbitration appears to reward traditional performance outcomes, although advanced contact-quality metrics also demonstrate predictive value
- Elite pitchers may receive premiums beyond measurable performance characteristics

LASSO Extension

- Takeaways
 - Cross-Validation used to select the penalty parameter (λ)
 - Simplest model within one standard error retained:
 - Quality Starts
 - xwOBA
 - Barrel Rate
 - RMSE = .239 (vs. .231 for OLS)
 - LASSO emphasized workload outcomes and contact-quality measures while shrinking traditional rate statistics toward zero

